Coincidence Classes in the ECDSA State Space: A Zero-Yield Theorem, with Exact Frequencies for Classes A, B and E

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Abstract

We analyse a finite combinatorial object: the state space of an idealised ECDSA signer, together with a family of subsets defined by exact arithmetic coincidences between the *hidden* pair $(s_{zk}, s_{rxk}) = (zk^{-1}, rxk^{-1})$ and quantities computable from public data. Three such subsets — the classes A, B and E of a previously circulated classification — are treated in closed form.

The organising result is a conservation law. Let R be *any* rule mapping the public data of a signature to a set C of candidate values for the hidden component. Then $\Pr[s_{zk} \in C] = \mathbb{E}|C|/(n-2)$ exactly, so the success probability per tested candidate is the constant $1/(n-2)$ for every rule, identical to sampling uniformly from $\mathbb{Z}_n^* \setminus \{s\}$. Classification cannot change the quality of candidates; it can change only their number. The classes are subsets, their frequencies are cardinalities, and nothing further is available.

Instantiating the theorem yields exact frequencies, proved and then confirmed by exhaustive enumeration of 1.66×10^9 states. Class A emits $\mathbb{E}|C| = \Pr[\mathcal{Q}] \to 1/4$ candidates per signature; class E emits $5/6$ and is shown to be *exactly* the affine locus $s_{zk} = 3s_{rxk} + 2$, whence $\Pr[\mathrm{E}] = 2\lfloor (n-3)/3 \rfloor/(n-1)^2 \sim 2/(3n)$; class B emits $\Theta(n^{-1} \log n)$, and its multiplier satisfies an *integer* quadratic, so no modular square root arises. Three structural facts follow: the qualification rate is exactly $1/4$ on every curve, being the area of a lattice region; the "Goldbach" offset distinguishing class E is one of $\Theta(n)$ interchangeable choices; and the recovery formulas of the classification are a single Möbius bijection in disguise.

On secp256k1 the frequencies are $2^{-258.00}$, $2^{-256.58}$ and $\approx 2^{-506.9}$ per signature. We report them because the question is natural, not because they are decisive: by the main theorem the entire framework improves on uniform random key guessing by a factor of exactly $1 + 1/(n-2) \approx 1 + 10^{-77}$, and that residual advantage is bought by the observation $s_{zk} \neq s$ rather than by any of the algebra. This is a negative result. Its value is that it closes a natural line of enquiry with proofs rather than with estimates.

Keywords: ECDSA · secp256k1 · lattice-point counting · negative result · hidden number problem · nonce distribution

MSC 2020: 11T71, 94A60, 11H06, 05A15

Statement of non-disclosure. This paper reports no vulnerability. It describes no attack, no partial attack and no weakening of any deployed parameter set. Its main theorem proves that a family of proposed techniques cannot outperform exhaustive key search, which is itself $\approx 2^{128}$ times more expensive than the generic attack the curve was designed to resist. Readers arriving from the earlier "vulnerability" framing of this project's own preprints are directed to Section 12, in which that framing is withdrawn.

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1 Introduction

1.1 Motivation

ECDSA signing computes $s = (z + rx)k^{-1} \bmod n$, and the scalar s splits additively into a message part and a key part,

$$s_{zk} = zk^{-1} \bmod n, \qquad s_{rxk} = rxk^{-1} \bmod n, \qquad s \equiv s_{zk} + s_{rxk} \pmod n.$$

Both summands are secret. An analyst with access to (z, r, s) can nevertheless form auxiliary integers — most naturally $s_{zr} = zr \bmod n$, the difference $d = s_{zr} - s$, and the remainder $a = s \bmod d$ — and ask whether the hidden summands happen to stand in an exact arithmetic relation to them. When they do, and when the relation pins s_{zk} to a small set of values, the private key follows immediately, because s_{zk} determines the nonce and the nonce determines the key.

Such coincidences do occur. On reduced-size test curves they occur often enough to be catalogued, and a previously circulated framework for this project catalogued five of them, labelled A through E, deriving closed-form key-recovery procedures for A, B and E. Empirically the classes thin out as the curve order n grows, and the natural questions are how fast, how far, and whether the thinning is a barrier or merely a nuisance.

This paper answers all three, and then shows that answering them was unnecessary.

1.2 Contributions

1. **A conservation law (Theorem 6).** For every rule computable from public data, the probability of a correct guess of s_{zk} equals the expected number of candidates emitted, divided by $n - 2$. The yield per candidate is therefore a constant, independent of the rule. No classification of ECDSA signatures — existing, proposed, or not yet conceived — can beat uniform random guessing on $\mathbb{Z}_n^* \setminus \{s\}$ (Corollary 7). This subsumes every frequency computation in the paper and closes the family.

2. **A structure theorem for class E (Theorem 22).** The three-clause "Goldbach-type" definition of class E is equivalent to the single affine condition $s_{zk} = 3s_{rxk} + 2$ over $\mathbb{Z}$. Its exact cardinality follows (Theorem 24), as does the fact that the distinguished offset is arbitrary (Theorem 26).

3. **Exact frequencies where the prior work had heuristics.** $\Pr[\mathcal{Q}] = 1/4 + O(n^{-1} \log n)$ (Theorem 11); $\Pr[\mathrm{A}] = \Pr[\mathcal{Q}]/(n-2)$, an identity (Theorem 15); $\Pr[\mathrm{E}] = 2\lfloor (n-3)/3 \rfloor/(n-1)^2$, an identity (Theorem 24); $\Pr[\mathrm{B}] = \Theta(n^{-2} \log n)$ (Proposition 19).

4. **Class B lives over $\mathbb{Z}$ (Theorem 17).** Its multiplier satisfies an integer quadratic whose discriminant must be a perfect square — a far stronger constraint than quadratic residuosity modulo n, and one that removes the modular square root from the published procedure. Wrap-around instances are shown to require $m \leq 2$ together with an exact auxiliary relation (Lemma 18).

5. **Verification.** Exhaustive enumeration of the full state space for seven primes (1.66×10^9 states), Monte-Carlo runs at $n \in \{9\,967,\ 99\,667\}$, and comparison against three independently generated test-curve datasets spanning five orders of magnitude in n. Every identity is confirmed to the precision of the enumeration; every asymptotic estimate is confirmed to within Poisson fluctuation.

6. **Errata (Section 12).** Three corrections to previously circulated material, one of them to an earlier draft of this paper. The most consequential is that the published class-E rate $1/(n-1)$ overstates the truth by 50% and was already excluded at 32σ by the data accompanying it.

1.3 What this paper does not claim

It is not a vulnerability disclosure. It contains formulas that map a *correct guess* of s_{zk} to the private key, and those formulas are correct; but by Proposition 3 they constitute a single Möbius bijection, that is, a change of coordinates on a search space of size $n - 1$, and by Theorem 6 no public-data rule can make the guess better than uniform. Reading them as an attack is a category error.

The practical irrelevance is not a matter of parameters being merely large. The obstruction is information-theoretic: conditioning on every public quantity leaves the conditional law of s_{zk} uniform on $n - 2$ values, so there is no leakage for any algorithm, present or future, to amplify.

1.4 Related work

Attacks that do recover ECDSA keys exploit a defect in the signer rather than a coincidence in the output. Repeated nonces yield the key from two signatures by elementary algebra and have been found repeatedly in the wild [3, 4]. Partially known or biased nonces reduce to the Hidden Number Problem of Boneh and Venkatesan [2] and are solved by lattice reduction [7, 10, 11]; a handful of leaked bits over a few dozen to a few million signatures suffices, as demonstrated against real deployments by Breitner and Heninger [4], the Minerva group [8], and LadderLeak [1], the last operating on less than one bit of leakage per signature. Deterministic nonce derivation [14] removes the randomness failure mode at the cost of introducing a fault-attack surface. The common feature is that each exploits $\Theta(\ell)$ bits of genuine information per signature; Theorem 6 shows the classes studied here supply zero, which is why they belong to a different category rather than to a weaker point on the same spectrum.

Generic attacks on the underlying discrete logarithm cost $\Theta(\sqrt{n}) \approx 2^{128}$ operations on secp256k1 [13]; the curve parameters are those of SEC 2 [5], and the signature scheme is standardised in FIPS 186-5 [12]. The counting arguments below use Dirichlet's divisor theorem in the form given by Hardy and Wright [6], and the physical bounds of Section 11 use Landauer's principle [9].

1.5 Organisation

Section 2 fixes notation and proves the master identity. Section 3 states the probability model and its structure lemma. Section 4 proves the main theorem; a reader interested only in the conclusion may stop there. Sections 5–8 instantiate it for the qualification event and for classes A, B, E. Section 9 reports the verification, Section 10 evaluates on secp256k1, Section 11 adds physical bounds, Section 12 records the errata, and Section 13 discusses falsifiability and limitations.

2 Preliminaries

2.1 Signatures and hidden components

Let $E/\mathbb{F}_p$ be an elliptic curve with base point G of prime order n, and let $Q = xG$ be the public key of $x \in \{1, \dots, n-1\}$. For a message hash z and nonce k, both in $\{1, \dots, n-1\}$,

$$r = \left[kG\right]_x \bmod n, \qquad s = (z + rx)k^{-1} \bmod n. \tag{1}$$

We write $\mathbb{Z}_n^* = \{1, \dots, n-1\}$ and identify every element of $\mathbb{Z}_n$ with its representative in $\{0, \dots, n-1\}$. This identification is essential, because the conditions below mix modular arithmetic with ordinary integer comparison, divisibility and order. Relations meant over $\mathbb{Z}$ rather than over $\mathbb{Z}_n$ are marked as such.

Definition 1 (Hidden components)**.** $\zeta := s_{zk} = zk^{-1} \bmod n$ and $\xi := s_{rxk} = rxk^{-1} \bmod n$, so that $s \equiv \zeta + \xi \pmod{n}$. Neither is observable from (z, r, s).

Since $\zeta, \xi \in \{1, \dots, n-1\}$, the *integer* sum $\sigma := \zeta + \xi$ lies in $\{2, \dots, 2n-2\}$ and satisfies exactly one of

$$\sigma = s \quad \text{(no wrap-around)}, \qquad \text{or} \qquad \sigma = s + n \quad \text{(wrap-around)}. \tag{2}$$

The distinction is invisible publicly and matters for classes B and E.

Definition 2 (Public auxiliaries)**.** $s_{zr} := zr \bmod n$; when $s_{zr} > s$, $d := s_{zr} - s$ and $a := s \bmod d$, the last an integer remainder. All are computable from the signature and the signed message.

2.2 The master identity

Proposition 3 (Master identity; recovery is a relabelling)**.** *(i) If $\hat{s} = \zeta$, then* $x = z(s - \hat{s})(r\hat{s})^{-1} \bmod n$.

(ii) The map $\varphi : \mathbb{Z}_n^* \setminus \{s\} \to \mathbb{Z}_n^*$, $\varphi(\hat{s}) = z(s - \hat{s})(r\hat{s})^{-1} \bmod n$, is injective. Hence knowing ζ is equivalent to knowing x, and the formula in (i) transports no information from public data to the secret.

Proof. (i) From Definition 1, $k = z\zeta^{-1}$ and $\xi = (s - \zeta) \bmod n$. Substituting into $\xi = rxk^{-1}$ gives $x = \xi k r^{-1} = (s - \zeta) \cdot z\zeta^{-1} \cdot r^{-1} = z(s - \zeta)(r\zeta)^{-1}$. (ii) Expand: $\varphi(\hat{s}) = zsr^{-1}\hat{s}^{-1} - zr^{-1}$. Inversion is a bijection of $\mathbb{Z}_n^*$, and $t \mapsto (zsr^{-1})t - zr^{-1}$ is an affine bijection since $zsr^{-1} \neq 0$ in the field $\mathbb{Z}_n$. A composition of bijections is injective. □

Proposition 3(ii) governs the subject. Searching for x through $\hat{s}$ is a change of coordinates on a space of size $n-1$. Every class below is therefore no more than *a rule for proposing candidate values of* ζ, and only two things about such a rule matter: how often the proposal is correct, and how many proposals it costs.

3 The probability model

Assumption A (Ideal signer). z, x and k are drawn independently and uniformly from $\mathbb{Z}_n^*$.

Assumption B (Abscissa heuristic). $r = [kG]_x \bmod n$ is modelled as uniform on $\mathbb{Z}_n^*$ and independent of (z, x, k).

Assumption A is the definition of a correctly implemented signer; when it fails, ECDSA is broken by the mechanisms of Section 1.4, which have nothing to do with this paper. Assumption B is a genuine heuristic, since r is a deterministic function of k. It is adopted only to obtain closed forms. Every prediction derived from it is checked in Section 9 against datasets generated on actual elliptic curves, where no such assumption was made; agreement is within Poisson fluctuation across five orders of magnitude in n and eight in sample size. Assumption B is therefore not load-bearing for the numerical conclusions.

Lemma 4 (Structure of the state). *Put $u := kr \bmod n$. Under Assumptions A and B:*

(i) (ζ, ξ, u) *is uniform on* $(\mathbb{Z}_n^*)^3$;

(ii) the map $(\zeta, \xi, u) \mapsto (\zeta, s, s_{zr})$ *with* $s = \zeta + \xi$ *and* $s_{zr} = \zeta u$ *(both mod n) is a bijection onto*

$$\mathcal{S} \;=\; \{(\zeta, s, s_{zr}) \;:\; \zeta \in \mathbb{Z}_n^*,\ s \in \mathbb{Z}_n \setminus \{\zeta\},\ s_{zr} \in \mathbb{Z}_n^*\}, \qquad |\mathcal{S}| = (n-1)^3,$$

so (ζ, s, s_{zr}) *is uniform on* $\mathcal{S}$;

(iii) consequently s_{zr} *is uniform on* $\mathbb{Z}_n^*$ *and independent of* (ζ, s), *and for each fixed* $s \neq 0$ *the conditional law of* ζ *given* (s, s_{zr}) *is uniform on* $\mathbb{Z}_n^* \setminus \{s\}$, *a set of exactly* $n-2$ *elements.*

Proof. (i) Fix k, r. Then $\zeta = zk^{-1}$ is a bijective image of the uniform z, hence uniform on $\mathbb{Z}_n^*$; $\xi = rxk^{-1}$ is a bijective image of the uniform x, hence uniform; they are independent because $z \perp x$. Neither depends on $u = kr$, a function of (k, r) alone; by Assumption B, u is uniform on $\mathbb{Z}_n^*$ and independent of (z, x), hence of (ζ, ξ). (ii) The map is invertible by $\xi = s - \zeta$, $u = s_{zr}\zeta^{-1}$, and its image is exactly $\mathcal{S}$, since $\xi \neq 0 \iff s \neq \zeta$ and $u \neq 0 \iff s_{zr} \neq 0$. A bijection carries the uniform law to the uniform law, and $|\mathcal{S}| = (n-1)^3$ because for each ζ there are $n-1$ admissible values of s. (iii) Immediate from the product structure of $\mathcal{S}$. $\square$

Lemma 4(iii) is the workhorse. After the analyst has computed everything computable — s, s_{zr}, d, a, and any function of them — the hidden component ζ remains uniform over $n-2$ possibilities. Conditioning on public data leaves the conditional entropy of the secret at $\log_2(n-2)$ bits.

4 The zero-yield theorem

Definition 5 (Public-data rule). A *rule* is a map R from the public data of a signature — the curve, z, r, s, and anything computable from them, such as s_{zr}, d, a — to a finite candidate set $C = R(z, r, s) \subseteq \mathbb{Z}_n^*$ for the hidden component ζ. Randomised rules are permitted, provided their randomness is independent of the signer's. The rule *succeeds* if $\zeta \in C$.

Every class of the classification is a rule. Class A emits $C = \{a\}$ on qualified transactions and $C = \varnothing$ otherwise. Class E emits the roots of (10) for each admissible branch of σ, discarding those outside $[1, n-1]$. Class B emits $\{am\}$ for each integer root m of (5). Uniform guessing is the rule that emits t i.i.d. uniform draws.

Theorem 6 (Zero yield). *For every rule R,*

$$\Pr[\zeta \in C] \;=\; \frac{\mathbb{E}\big|\, C \cap (\mathbb{Z}_n^* \setminus \{s\})\,\big|}{n-2} \;\leq\; \frac{\mathbb{E}|C|}{n-2}. \tag{3}$$

In particular the success probability per candidate emitted *is the constant* $1/(n-2)$, *independent of R.*

Proof. By Lemma 4(ii) the triple (ζ, s, s_{zr}) is uniform on the product set $\mathcal{S}$. Hence, conditionally on the public data — a function of (s, s_{zr}, r), all independent of ζ by Lemma 4(iii) — the hidden component ζ is uniform on $\mathbb{Z}_n^* \setminus \{s\}$, a set of exactly $n-2$ elements. The set C is measurable with respect to that same public data together with auxiliary randomness independent of the signer, hence conditionally fixed. For a uniform variable on a set of size $n-2$ and a fixed subset C, $\Pr[\zeta \in C \mid \text{public}] = |C \cap (\mathbb{Z}_n^* \setminus \{s\})|/(n-2)$. Take expectations. $\square$

Corollary 7 (Universal no-go)**.** *No public-data classification of ECDSA signatures can achieve a per-candidate success rate different from* $1/(n-2)$*. Consequently:*

(i) no rule is better *than uniform guessing on* $\mathbb{Z}_n^* \setminus \{s\}$*; equality in* (3) *holds for that rule and for every rule emitting distinct candidates;*

(ii) a rule can be worse *only by emitting duplicates or the excluded value* s*, both implementation defects rather than properties of the class;*

(iii) against a guesser drawing uniformly from all of $\mathbb{Z}_n^*$*, whose yield is* $1/(n-1)$*, the best possible rule wins by the factor* $(n-1)/(n-2) = 1 + 1/(n-2)$*, equal to* $1 + 8.64 \times 10^{-78}$ *on secp256k1. That advantage is purchased entirely by the observation* $\zeta \neq s$*, which requires no algebra.*

Remark 8. The correct negative statement is an equality, not an inequality. “Worse than guessing” would suggest an inefficiency that a cleverer rule might remove; “exactly equal to guessing, for every rule” is a conservation law, and there is nothing left to optimise. Sections 5 through 8 may therefore be read as a computation of the single scalar $\mathbb{E}|C|$ for three particular rules, undertaken because the classes were already in circulation and their frequencies had been misreported.

5 The qualification event

Definition 9 (Qualification)**.** A transaction *qualifies*, written $\mathcal{Q}$, if $s_{zr} > s$ and the residue $a = s \bmod d$ satisfies $a > 0$, $a \neq s$ and $a \neq s_{zr}$.

Lemma 10 (Geometric form)**.** *$\mathcal{Q}$ holds if and only if $s < s_{zr} \leq 2s$ and $d \nmid s$.*

Proof. Assume $s_{zr} > s$, so $d \geq 1$. If $d > s$ then $s \bmod d = s$, violating $a \neq s$; conversely if $d \leq s$ then $a < d \leq s$, so $a \neq s$. Thus $a \neq s \iff d \leq s \iff s_{zr} \leq 2s$. Next, $a = 0 \iff d \mid s$. Finally $a < d < s_{zr}$ always, so $a \neq s_{zr}$ is automatic. $\square$

Theorem 11 (Qualification rate)**.** $\Pr[\mathcal{Q}] = \frac{1}{4} - \Theta\big(n^{-1}\log n\big) \to \frac{1}{4}$*, independently of the curve, the field and the key.*

Proof. By Lemma 4(iii) the pair (s, s_{zr}) is, up to an $O(1/n)$ perturbation of the marginal of s, uniform on $\{1, \ldots, n-1\}^2$. Write $M = n-1$, which is even for every odd prime n, and count lattice points with $s < s_{zr} \leq \min(2s, M)$:

$$\sum_{s=1}^{M}\big(\min(2s, M) - s\big) = \underbrace{\sum_{s=1}^{M/2} s}_{=M^2/8+M/4} + \underbrace{\sum_{s=M/2+1}^{M} (M-s)}_{=M^2/8-M/4} = \frac{M^2}{4},$$

an exact identity. The excluded set $d \mid s$ contributes $\sum_{s\leq M} \tau(s) = M\log M + (2\gamma-1)M + O(\sqrt{M})$ points by Dirichlet’s divisor theorem [6, Thm. 320]. Dividing by M^2 gives $\Pr[\mathcal{Q}] = \frac{1}{4} - (\log M + 2\gamma - 1)/M + O(M^{-1/2})$, and the perturbation of the marginal of s adds a further $O(1/M)$. $\square$

Remark 12. The often-quoted “25% anomaly rate” of this project is thus the area of the region $s < s_{zr} \leq 2s$ in the unit square. It is scale-invariant because that area is $1/4$ for every n. It measures the geometry of comparing two independent residues and carries no information about key recoverability; its constancy across curves is a reason to expect no exploitability, not a reason to hope for it.

6 Class A: linear divisibility with unit multiplier

Definition 13 (Class A)**.** A qualified transaction is in class A if the ratio $m_1 = \zeta/a$, computed in $\mathbb{Q}$, equals 1; equivalently $\zeta = a$ as integers. (The strict taxonomy additionally demands $m_2 = (s + s_{zr})/\xi \notin \mathbb{Z}$, separating A from B; by Proposition 19 that refinement removes an $O(n^{-2}\log n)$ fraction and is invisible at every scale considered here.)

Theorem 14 (Recovery)**.** *On a class-*A *transaction,* $k = az^{-1} \bmod n$ *and* $x = (s-a)(rk)^{-1} = z(s-a)(ra)^{-1} \bmod n$*. One candidate is produced, correct with certainty, verified by one scalar multiplication* $xG \stackrel{?}{=} Q$*.*

Proof. Substitute $\hat{s} = a = \zeta$ into Proposition 3(i). All inverses exist because n is prime and $z, r, a \not\equiv 0$. □

Theorem 15 (Exact frequency)**.** *For every prime* $n \geq 5$*,*

$$\Pr[\mathrm{A}] \;=\; \frac{\Pr[\mathcal{Q}]}{n-2} \;=\; \frac{1}{4(n-2)}\Big(1 - \Theta\big(n^{-1}\log n\big)\Big) \;\sim\; \frac{1}{4n}, \tag{4}$$

*an identity in the model of Lemma 4, not an asymptotic estimate. Equivalently, the class-*A *rule emits* $\mathbb{E}|C| = \Pr[\mathcal{Q}]$ *candidates per signature.*

Proof. Condition on (s, s_{zr}). If it does not qualify, $C = \varnothing$. If it qualifies then d and a are determined, and by Lemma 10 $1 \leq a < d \leq s$, so $a \in \mathbb{Z}_n^* \setminus \{s\}$ and $|C| = 1$. Apply Theorem 6. □

Erratum 1 (A wrong turn worth recording). It is tempting to compute $\Pr[\mathrm{A}]$ by counting, for each (s, ζ), the divisors $d \mid (s - \zeta)$ with $d > \zeta$, and approximating the resulting sum by an integral. That route yields $\Pr[\mathrm{A}] \approx 0.3466/n$, which is 39% too large and is rejected by the tiny-curve data at 16.8σ. The continuum approximation silently discards the fractional parts $\{(n-\zeta)/d\}$, whose systematic mean of $\frac{1}{2}$ accumulates to a term of order d under the ζ-summation. Carrying the floors exactly gives $\sum_{\zeta=0}^{d-1} \lfloor (M-\zeta)/d \rfloor = M - d + 1$ and restores $M^2/4$, in agreement with Theorem 15. Lattice counts in this subject are dominated by their floor terms; integral approximations to them should be distrusted by default.

7 Class B: dual integral divisibility

Definition 16 (Class B)**.** A qualified transaction is in class B if $m_1 = \zeta/a$ and $m_2 = (s + s_{zr})/\xi$ are both integers and $m_1 = m_2 =: m$. Both quotients are taken in $\mathbb{Q}$: they assert exact divisibility over $\mathbb{Z}$, not congruences.

Theorem 17 (The multiplier satisfies an integer quadratic)**.** *On a class-*B *transaction, with* $\sigma \in \{s,\, s+n\}$ *the true integer sum* (2)*,*

$$a\,m^2 \;-\; \sigma\,m \;+\; (s + s_{zr}) \;=\; 0 \qquad \text{in } \mathbb{Z}. \tag{5}$$

Hence $\Delta = \sigma^2 - 4a(s + s_{zr})$ *is a perfect square in* $\mathbb{Z}$ *and* $m = (\sigma \pm \sqrt{\Delta})/(2a) \in \mathbb{Z}$*, giving* $\zeta = am$ *and* $x = z(s-\zeta)(r\zeta)^{-1} \bmod n$*. At most four candidates arise, two roots for each branch of* σ*.*

Proof. By definition $\zeta = am$ and $s + s_{zr} = m\xi$ over $\mathbb{Z}$. By (2), $\xi = \sigma - \zeta = \sigma - am$, so $s + s_{zr} = m(\sigma - am) = \sigma m - am^2$, which is (5). All coefficients are integers, so the quadratic formula over $\mathbb{Z}$ forces Δ to be a perfect square. □

Lemma 18 (Wrap-around in class B)**.** *If a class-*B *transaction has* $\sigma = s + n$*, then* $m \leq 2$*; moreover* $m = 1$ *implies* $s_{zr} + a = n$*, and* $m = 2$ *implies that* d *is even and* $4a + d = 2n$*.*

Proof. Qualification gives $d \leq s$, so $s + s_{zr} = 2s + d \leq 3s$. Also $\xi = s + n - \zeta \geq s + 1$ because $\zeta \leq n - 1$. From $m\xi = s + s_{zr}$ we get $m \leq 3s/(s+1) < 3$. For $m = 1$: $\xi = 2s + d$ and $\xi = s + n - a$ give $n = s + d + a = s_{zr} + a$. For $m = 2$: $2\xi = 2s + d$ forces d even with $\xi = s + d/2$, while $\xi = s + n - 2a$; equating gives $2n = 4a + d$. □

Each is one exact equation between quantities of size $\Theta(n)$, hence a codimension-one condition on top of an already $O(n^{-2})$-rare event. No wrap-around instance occurred in the exhaustive enumeration of Section 9. Implementations should nonetheless test both branches; the cost is one extra scalar multiplication.

Proposition 19 (Frequency of class B)**.** *Under a divisor-equidistribution heuristic,*

$$\Pr[\mathrm{B}] = \Theta\left(\frac{\log n}{n^2}\right), \qquad \textit{empirically} \qquad \Pr[\mathrm{B}] \approx \frac{c_B \log n}{n^2}, \quad c_B \approx 0.19. \tag{6}$$

Argument. Fix (s, s_{zr}) qualified. Each divisor m of $s + s_{zr}$ determines a unique candidate pair $\zeta = am$, $\xi = (s + s_{zr})/m$; the pair survives only if additionally $\zeta + \xi \in \{s,\, s+n\}$, a coincidence of probability $\Theta(1/n)$ under equidistribution. The mean of τ on $[1, 2n]$ is $\log n + O(1)$, so the conditional rate is $\Theta(n^{-1} \log n) \cdot n^{-1}$; multiplying by $\Pr[\mathcal{Q}] = \frac{1}{4}$ gives $\approx \frac{1}{4} n^{-2} \log n$. Exhaustive enumeration for $101 \le n \le 1009$ yields $\Pr[\mathrm{B}]\, n^2/\log n \in [0.189,\, 0.196]$, so $c_B \approx 0.19$, within 25% of the heuristic constant $\frac{1}{4}$. □

Remark 20 (Weakest link, stated plainly). The measured ratio drifts downward by about 3% across the accessible range, so the true law may carry a sub-logarithmic correction that the data cannot resolve; the accessible range spans one decade of n while the extrapolation target spans 74. This is the least secure claim in the paper. It is also inconsequential: an error of two orders of magnitude in c_B moves $\Pr[\mathrm{B}]$ on secp256k1 from 10^{-153} to somewhere between 10^{-155} and 10^{-151}, and Theorem 6 is unaffected either way. Classes A and E carry no such caveat: their formulas are exact for every prime n.

8 Class E: the "Goldbach-type" decomposition

Class E is the most instructive of the three: its published description is ornate and its actual content is one line.

Definition 21 (Classes E1 and E2)**.** With $\sigma = \zeta + \xi$ the integer sum (2), a transaction is in class E if σ is even and

$$|\zeta - \xi| \;=\; \frac{\sigma}{2} + 1. \tag{7}$$

Instances with $\sigma = s$ are labelled E1 and those with $\sigma = s + n$ are labelled E2. The published formulation writes $\lfloor \sigma/2 \rfloor + 1$; since σ is required to be even, the floor is vacuous.

Theorem 22 (Class E is an affine line)**.** *For integers $\zeta, \xi \ge 1$, condition* (7) *together with "σ even" holds if and only if*

$$\zeta = 3\xi + 2 \qquad \textit{or} \qquad \xi = 3\zeta + 2 \qquad \textit{in } \mathbb{Z}. \tag{8}$$

*Moreover every class-*E *instance satisfies $\sigma \equiv 2 \pmod 4$, and the two branches are disjoint.*

Proof. ($\Rightarrow$) Equality (7) rules out $\zeta = \xi$, its right-hand side being positive. Take $\zeta > \xi$ without loss of generality; then $\zeta - \xi = (\zeta + \xi)/2 + 1$, i.e. $2\zeta - 2\xi = \zeta + \xi + 2$, i.e. $\zeta = 3\xi + 2$. ($\Leftarrow$) If $\zeta = 3\xi + 2$ then $\sigma = 4\xi + 2$ is even and $\zeta - \xi = 2\xi + 2 = \sigma/2 + 1$, so (7) holds; and $\sigma = 4\xi + 2$ gives $\sigma \equiv 2 \pmod 4$. If both branches held, $\zeta = 3(3\zeta + 2) + 2 = 9\zeta + 8$ forces $\zeta = -1$, impossible for a positive integer. □

Remark 23 (The Goldbach framing is empty). Theorem 22 dissolves the name. The published description — "a symmetric pair around $\sigma/2$ with fixed difference" — is doubly vacuous: *every* pair summing to σ is symmetric about $\sigma/2$ by construction, and the only content of (7) is that the hidden pair lies on one particular line of slope 3. There is no additive number theory here and no connection to the Goldbach problem beyond the superficial fact that a number is written as a sum of two others.

Theorem 24 (Exact cardinality and frequency)**.** *For every prime $n \ge 5$ the number of class-*E *hidden pairs in $\{1, \ldots, n-1\}^2$ is*

$$\#\mathrm{E} \;=\; 2\left\lfloor \frac{n-3}{3} \right\rfloor, \qquad \textit{so} \qquad \Pr[\mathrm{E}] \;=\; \frac{2\lfloor (n-3)/3 \rfloor}{(n-1)^2} \;=\; \frac{2}{3n} + O\left(\frac{1}{n^2}\right). \tag{9}$$

Proof. By Theorem 22 the class is a disjoint union of two branches. On $\zeta = 3\xi + 2$ the constraints are $\xi \geq 1$ and $3\xi + 2 \leq n - 1$, i.e. $1 \leq \xi \leq (n-3)/3$, giving $\lfloor (n-3)/3 \rfloor$ solutions; the other branch gives the same count by symmetry. Class E depends only on (ζ, ξ), uniform on $(\mathbb{Z}_n^*)^2$ by Lemma 4(i), a set of $(n-1)^2$ elements. No class-E pair is removed by $s \neq 0$, since $\sigma = 4\xi + 2 = n$ would force n even. $\square$

Theorem 25 (Recovery)**.** *Given the effective sum $\sigma \in \{s, s+n\}$ with $\sigma \equiv 2 \pmod 4$, put*

$$\alpha = \frac{3\sigma + 2}{4}, \qquad \beta = \sigma - \alpha = \frac{\sigma - 2}{4}. \tag{10}$$

Then $\{\zeta, \xi\} = \{\alpha, \beta\}$, and x is one of $x_c = z(s-c)(rc)^{-1} \bmod n$ for $c \in \{\alpha, \beta\}$, disambiguated by one scalar multiplication.

Proof. Solve $\zeta + \xi = \sigma$, $\zeta - \xi = \sigma/2 + 1$: adding gives $2\zeta = 3\sigma/2 + 1$, i.e. $\zeta = (3\sigma + 2)/4$, an integer precisely because $\sigma \equiv 2 \pmod 4$ (Theorem 22). Apply Proposition 3(i) to each candidate. $\square$

Theorem 26 (Offset invariance)**.** *For any fixed $c \in \mathbb{Z}$ define E_c by "σ even and $|\zeta - \xi| = \sigma/2 + c$". Then $\mathrm{E}_c = \{\zeta = 3\xi + 2c\} \sqcup \{\xi = 3\zeta + 2c\}$, and for $|c| = o(n)$,*

$$\Pr[\mathrm{E}_c] = \frac{2}{3n}\Big(1 + O\big(|c|/n\big)\Big),$$

independently of c. The Goldbach value $c = 1$ is thus one member of a family of $\Theta(n)$ equally frequent classes.

Proof. Repeat Theorem 22 with right-hand side $\sigma/2 + c$: for $\zeta > \xi$, $2\zeta - 2\xi = \zeta + \xi + 2c$ gives $\zeta = 3\xi + 2c$; conversely $\sigma = 4\xi + 2c$ is even and $\zeta - \xi = 2\xi + 2c = \sigma/2 + c$. Counting on one branch, $\#\{\xi : 1 \leq \xi \leq n-1,\ 1 \leq 3\xi + 2c \leq n-1\} = n/3 + O(|c|)$; the other is symmetric. $\square$

A control experiment in the project repository reports a flat frequency of $\approx 0.67/n$ for offsets $c \in \{\pm 1, 3, 5, 7, 9, 101\}$. Theorem 26 accounts for it exactly: the constant $0.67 \approx 2/3$ is $\Pr[\mathrm{E}_c] \cdot n$, and its independence of c is a two-line consequence of the affine structure. An experiment run to test a hypothesis turns out to be a corollary of that hypothesis's own algebra.

8.1 Class versus detector

The rule of Definition 5 associated with class E must try both branches of σ, since it cannot see which is real. Counting admissible candidates: $\sigma = s$ contributes two whenever $s \equiv 2 \pmod 4$; $\sigma = s + n$ contributes two when additionally $s \leq n/3$ and one otherwise, because α then exceeds $n - 1$ while β does not. Hence

$$\mathbb{E}|C_{\mathrm{E}}| = \tfrac{1}{4} \cdot 2 + \tfrac{1}{4}\Big(\tfrac{1}{3} \cdot 2 + \tfrac{2}{3} \cdot 1\Big) = \tfrac{5}{6}, \qquad \Pr[\text{rule E succeeds}] = \frac{5}{6(n-2)}, \tag{11}$$

which exceeds $\Pr[\mathrm{E}] = 2/(3n)$ of Theorem 24. There is no contradiction: the *class* is defined using the secret σ, whereas the *rule* also scores accidental hits from the wrong branch, and the gap $5/6$ against $2/3$ is exactly their weight. This distinction — between a class defined with oracle access to the secret and a rule that must operate without it — is where informal treatments of the subject usually go wrong.

9 Numerical validation

9.1 Exhaustive enumeration

For small n the entire state space $\mathcal{S}$ of Lemma 4, of size $(n-1)^3$, can be enumerated without sampling, giving exact probabilities. Table 1 reports all states for seven primes, 1.66×10^9 in total.

9.2 Independently generated test-curve data

The datasets accompanying the earlier framework were generated on genuine elliptic curves, so Assumption B played no role in producing them. Table 2 compares those counts with the formulas proved above. All residuals lie within Poisson fluctuation; the largest, $+2.05\sigma$ on 12 observed events, has a one-sided p-value of 0.05. Measured qualification rates were 0.2494 and 0.2503 on the two million-transaction datasets, against $1/4$ predicted.

n	states	$\Pr[\mathcal{Q}]$	$\Pr[\mathrm{A}]$	$\Pr[\mathcal{Q}]/(n-2)$	#E	$2\lfloor\frac{n-3}{3}\rfloor$	$\Pr[\mathrm{B}]n^2/\ln n$
101	1.00e6	0.2096820	2.118000e-3	2.118000e-3	64	64	0.192
151	3.38e6	0.2205200	1.480000e-3	1.480000e-3	98	98	0.195
251	1.56e7	0.2303389	9.250560e-4	9.250560e-4	164	164	0.196
401	6.40e7	0.2364823	5.926875e-4	5.926875e-4	264	264	0.193
503	1.26e8	0.2388253	4.766967e-4	4.766967e-4	332	332	0.194
751	4.22e8	0.2419628	3.230483e-4	3.230483e-4	498	498	0.191
1009	1.03e9	0.2437219	2.420277e-4	2.420277e-4	670	670	0.189

Table 1: Exhaustive enumeration of the full state space. Columns 4–5 confirm Theorem 15 as an identity to every displayed digit; columns 6–7 confirm (9) exactly; the last column calibrates $c_B \approx 0.19$ in (6). $\Pr[\mathcal{Q}]$ approaches $1/4$ from below at the rate $\Theta(\log n/n)$ of Theorem 11.

curve	n	N_{tx}	A pred	A obs	z_{A}	B pred	B obs	E pred	E obs	z_{E}
tiny	9 967	9.93e7	2 491.8	2 465	−0.54	1.75	2	6 642.2	6 746	+1.27
small	99 667	1.00e6	2.51	4	+0.94	2e-4	0	6.69	12	+2.05
large	1 241 630 743	1.00e6	2e-4	0	−0.01	3e-12	0	5e-4	0	−0.02

Table 2: Predicted versus observed counts on the project's test curves.

9.3 Monte-Carlo cross-check and the yield constant

Simulating the model of Lemma 4 with 2.4×10^7 draws per curve gave, at $n = 9\,967$: qualification rate 0.24909; #A $= 600$ against 599.8 predicted; #E $= 1\,614$ against 1 605.3. At $n = 99\,667$: qualification 0.24978, #A $= 71$ (predicted 60.1), #E $= 177$ (predicted 160.5). In all 4.8×10^7 draws the number of transactions satisfying Definition 21 was *identical* to the number satisfying the affine criterion (8) — $1\,614 = 1\,614$ and $177 = 177$ — confirming Theorem 22 without exception.

Table 3 tests Theorem 6 directly by measuring, for each rule, both the expected candidate count and the empirical yield per candidate.

10 Evaluation on secp256k1

The group order is the prime

$$n = \texttt{FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFE BAAEDCE6 AF48A03B BFD25E8C D0364141}_{16}$$

$$\approx 1.157920892 \times 10^{77},$$

with $\log_2 n = 256.000000$, $\ln n = 177.4457$, $n \equiv 1 \pmod 3$ and $n \equiv 1 \pmod 4$. Because $n \equiv 1 \pmod 3$ the floor in (9) evaluates exactly to $(n-4)/3$, so no approximation enters the class-A or class-E figures.

Taking $N_{\mathrm{tx}} = 10^{10}$ as a generous upper bound on all secp256k1 signatures ever produced — Bitcoin has processed on the order of 10^9 transactions carrying a few $\times 10^9$ input signatures, Ethereum a few $\times 10^9$ transactions — the expected number of instances of all three classes combined is 7.9×10^{-68}; at $N_{\mathrm{tx}} = 10^{12}$ it is 7.9×10^{-66}, and at an absurd 10^{20} it is 7.9×10^{-58}. Since these expectations are far below 1, Markov's inequality makes them upper bounds on the probability that even one such signature exists.

The sample size for even odds of a single occurrence is $N_{1/2} = \ln 2/\Pr$: 1.20×10^{77} for E, 3.21×10^{77} for A, 8.76×10^{76} for the union and 2.76×10^{152} for B. Every entry exceeds the group order.

Abundance without density. The classes are not empty on secp256k1. The transaction space $\mathcal{T} = \{1, \ldots, n-1\}^4$ has $|\mathcal{T}| \approx 1.80 \times 10^{308}$ elements, of which #A $\approx 3.88 \times 10^{230}$, #E $\approx 1.04 \times 10^{231}$ and #B $\approx 4.5 \times 10^{155}$. These are enormous absolute counts and a fraction $\approx 10^{-77}$ of the space. An analyst who reasons "there are 10^{231} vulnerable signatures out there" has said something true and entirely without consequence, in the same way that there are 10^{231} private keys beginning with any particular 256-bit prefix. Absolute counts in this subject should never be quoted without the density beside them.

rule	$\mathbb{E}\lvert C\rvert$ predicted	$\mathbb{E}\lvert C\rvert$ measured	yield/candidate measured	$1/(n-2)$
class A	$\Pr[\mathcal{Q}] \to 1/4$	0.2489	1.031×10^{-4}	1.004×10^{-4}
class E	$5/6$	0.8325	9.809×10^{-5}	1.004×10^{-4}
class B	$\approx 0.19 \ln n/n$	$0.184 \ln n/n$	—	1.004×10^{-4}

Table 3: Direct test of Theorem 6 at $n = 9\,967$ over 3×10^6 simulated signatures. The yield per candidate is the same constant for every rule, as the theorem requires; class B emitted too few candidates at this sample size for a yield estimate.

event	closed form	value	power of 2	status
$\mathcal{Q}$	$\frac{1}{4} + O(n^{-1} \log n)$	0.250000...	$2^{-2.000}$	theorem
A	$\Pr[\mathcal{Q}]/(n-2)$	2.159042e-78	$2^{-258.00}$	exact identity
B	$\approx 0.19 \ln n/n^2$	≈2.5e-153	$\approx 2^{-506.9}$	calibrated heuristic
E	$2\lfloor \frac{n-3}{3} \rfloor/(n-1)^2$	5.757446e-78	$2^{-256.58}$	exact identity
$\mathrm{A} \cup \mathrm{B} \cup \mathrm{E}$	union bound	7.916488e-78	$2^{-256.24}$	—
one blind guess	$1/(n-1)$	8.636169e-78	$2^{-256.00}$	exact

Table 4: Per-signature probabilities on secp256k1. These are per-signature figures; the like-for-like comparison at equal numbers of tested candidates is Corollary 7.

11 Physical bounds

Independently of any cryptanalytic comparison, the sample sizes above can be priced against physics. Landauer's principle [9] sets the minimum energy dissipated by an irreversible bit operation at $k_B T \ln 2$; at the cosmic microwave background temperature $T = 2.725\,\mathrm{K}$ this is $2.61 \times 10^{-23}\,\mathrm{J}$. Charging one bit erasure per trial — absurdly optimistic, since an ECDSA verification is 10^4 to 10^5 bit operations — enumerating $1/\Pr[\mathrm{A}] = 4.63 \times 10^{77}$ trials costs at least $1.21 \times 10^{55}\,\mathrm{J}$. That is the total mass–energy of 6.8×10^7 solar masses, or 2.0×10^{34} years of world primary energy consumption. The corresponding figure for class B is $1.04 \times 10^{130}\,\mathrm{J}$, some 10^{60} times the mass–energy of the observable universe.

The class-A figure deserves emphasis because it is the cheap one. Converting sixty-eight million stars entirely into computation, without losses, is the discounted price of encountering a single instance of the most common of the three classes — and by Corollary 7 the instance would yield one private key, obtainable with marginally better odds by guessing.

12 Errata to previously circulated material

This project circulated earlier write-ups of classes A and B under a "vulnerability" framing, together with a framework document stating frequency formulas for all five classes. That framing is withdrawn, and three specific numerical or structural claims are corrected here.

Erratum 2 (The class-E rate is $2/(3n)$, not $1/(n-1)$). The framework document states $\Pr[\mathrm{E}] = 1/(n-1)$, reasoning that the condition "fixes s_{zk} to at most two values" and that "exactly one of $s, s+n$ is even". The second clause is correct; the first over-counts. By Theorem 22 an even σ admits a solution only when $\sigma \equiv 2 \pmod 4$, which removes half the even sums, and the surviving locus has length $\approx n/3$ per branch rather than $\approx n/2$. The correct rate is $2/(3n)$, that is, $\frac{2}{3}$ of the published value.

The distinction is not academic. With $n = 9\,967$ and $N_{\mathrm{tx}} = 99\,323\,922$ — the project's own tiny-curve dataset, published alongside the formula — the model $1/(n-1)$ predicts 9 966.5 events against 6 746 observed, a Poisson deviation of -32.3σ; Theorem 24 predicts 6 642.2, a deviation of $+1.27\sigma$. The published formula was excluded by the data accompanying it.

Erratum 3 (No modular square root is required in class B). The published statement of the class-B recovery reduces the quadratic modulo n, computes $\sqrt{\Delta} \bmod n$ via the Euler criterion, and inverts $2a$ modulo n. That procedure is not incorrect, but it is strictly weaker and needlessly expensive: it discards the fact that (5) holds over $\mathbb{Z}$, where "Δ is a perfect square" is far more restrictive than "Δ is a quadratic residue mod n" (density $\Theta(\Delta^{-1/2})$ against $1/2$). It also fixes $\sigma = s$, silently assuming no wrap-around; see Lemma 18. In exhaustive enumeration for $n \leq 1009$ — 3 823 class-B instances — the integer discriminant was a perfect square in every case.

Erratum 4 (Correction to an earlier draft of this paper). An earlier draft compared an "anomaly hunter" processing T signatures against a guesser making $3T$ guesses, concluding that the hunter loses by a factor of 3.3. The arithmetic is right but the accounting is arbitrary: it charges the hunter for three candidates per signature, whereas the classes emit $\mathbb{E}|C| \approx \frac{1}{4} + \frac{5}{6} + O(n^{-1} \log n) \approx 1.083$. Charging both adversaries the same number of scalar multiplications gives an *exact tie*, up to the factor $1 + 1/(n-2)$ of Corollary 7. The corrected statement is stronger, for the reason given in Remark 8.

13 Discussion

13.1 Falsifiability

The results are conditional on Assumptions A and B. The useful question is what observation would overturn them. A class whose observed frequency exceeds its predicted c/n baseline by a factor $\lambda > 1$ that persists as n grows would demonstrate a genuine non-uniformity in the law of ζ — the only finding in this area worth pursuing, since it would contradict Lemma 4 and hence Theorem 6.

Such a test is cheap on test curves. For a Poisson count with baseline rate p_0, distinguishing p_0 from λp_0 at significance α and power $1 - \beta$ requires

$$N \gtrsim \frac{\left(z_{\alpha/2} + z_\beta\right)^2}{p_0\left(\sqrt{\lambda} - 1\right)^2}. \tag{12}$$

At $\alpha = 0.05$ and 80% power, detecting $\lambda = 2$ in class E on the tiny curve needs 6.8×10^5 transactions; $\lambda = 1.10$ needs 4.9×10^7; $\lambda = 1.01$ needs 4.7×10^9. The existing 9.9×10^7-transaction dataset already resolves a 10% enrichment.

Applying (12) to the data in hand: the tiny-curve class-E count (6 746 against 6 642.2 predicted) constrains λ to 1.016 ± 0.012, excluding any enrichment above 4% at 95% confidence; the class-A count (2 465 against 2 491.8) constrains λ to 0.989 ± 0.020. Both are consistent with $\lambda = 1$ to within a couple of percent.

Remark 27 (The one experiment worth running). Searching for further coincidence classes is not it: the space of arithmetic coincidences is infinite and, by Theorem 6, each new class will have frequency $\mathbb{E}|C|/(n-2)$ with $\mathbb{E}|C| = \Theta(1)$ at best. The informative experiment is a direct test of the *distribution*: take 10^8 to 10^9 transactions on a curve with $n \approx 10^4$–10^5, form the empirical law of $\zeta = zk^{-1}$ conditioned on every public statistic available, and test uniformity by a chi-squared statistic over $\sqrt{N}$ bins or by a discrete Fourier test in the manner of Bleichenbacher. A null result closes the question; a positive one would announce itself as a bias in a histogram, not as an elegant identity.

13.2 Limitations

1. **Assumption B is a heuristic.** Treating r as an independent uniform unit is not provable with current techniques. Its role is confined to obtaining closed forms, and the predictions were checked against data generated without it (Table 2).

2. **The class-B constant is empirical.** $c_B \approx 0.19$ is calibrated over $101 \leq n \leq 1009$ and drifts by $\approx 3\%$ across that range. Only the order $\Theta(n^{-2} \log n)$ is argued from first principles, and that rests on divisor equidistribution.

3. **Ideal signer only.** Nothing here bears on implementations with faulty randomness, deterministic-nonce bugs, fault injection or side channels; those are the real attack surface and are governed by the works cited in Section 1.4.

4. **Union bounds.** The three classes share the same hidden pair and are not independent, so the union figures in Table 4 are upper bounds — the conservative direction.

5. **Classes C and D are out of scope.** Class C admits no closed-form recovery. Class D is not a coincidence class but the generic case, occurring for $\approx$ 25% of transactions by Theorem 11; its candidate window has width $a = \Theta(n)$ on average, so $\mathbb{E}|C_D| = \Theta(n)$ and Theorem 6 gives success

probability $\Theta(1)$ at cost $\Theta(n)$ — that is, exhaustive search, roughly 2^{128} times worse than Pollard rho. The level index $N = \lfloor \zeta/a \rfloor$ is moreover a function of the secret ζ, so the window cannot be located without already knowing what it is meant to reveal.

13.3 On presentation

The results belong, if anywhere, as a short technical note in elementary combinatorics and its verification: exact cardinalities of a family of subsets of a finite product set, plus a conservation law that renders the family uninteresting. They do not belong in a disclosure channel, nor under a title containing the words "recovery", "vulnerability" or "attack" — not because the mathematics is weak, but because such a title would be read as a claim the mathematics refutes. Three specific cautions follow. The recovery formulas should appear only as corollaries of Proposition 3, never as procedures, since as procedures they require an input equivalent to the answer. The phrase "25% of transactions" should never appear without Theorem 11 attached, because in isolation it reads as an exposure rate when it is the area of a triangle. And absolute counts such as "10^{231} instances" should be reported only alongside the density 10^{-77}.

14 Conclusion

Under an ideal signer the hidden pair is uniform on $(\mathbb{Z}_n^*)^2$ and, jointly with the public s_{zr}, uniform on a product set. Conditioning on all public data therefore leaves ζ uniform over $n-2$ values, and every subsequent result is a consequence of that one sentence: the yield per tested candidate is the constant $1/(n-2)$ for every rule computable from public data, so a classification can change only how many candidates are emitted, never their quality. The best conceivable rule of this type beats uniform random key guessing by the factor $1 + 8.6 \times 10^{-78}$, and exhaustive key search is itself $\approx 2^{128}$ times more expensive than the generic attack secp256k1 was designed to resist.

Within that frame the three classes are exercises. Class A is the graph of a function and occurs with probability $\Pr[\mathcal{Q}]/(n-2)$; class E is the affine line $\zeta = 3\xi + 2$ and occurs with probability $2\lfloor (n-3)/3 \rfloor/(n-1)^2$; class B is a divisor coincidence of order $n^{-2}\log n$ whose multiplier satisfies an integer quadratic. The qualification rate is the area of a lattice region and equals $1/4$ on every curve. The distinguished offset of class E is one of $\Theta(n)$ equivalent choices.

The productive question is not whether more classes exist — infinitely many do, all of the same quality — but whether the distribution of s_{zk} is uniform. That question is answerable with existing data and existing tools, and the answer so far, to within $\pm 2\%$, is yes.

Data and code availability

All figures are reproduced by the enumeration in Appendix A, which is exhaustive for $n \leq 1009$ and therefore yields exact rationals rather than estimates. Datasets, the generator, and the control experiment are in the project repository at `[repositoryURL]`.

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A Reproduction code

The following is self-contained. For $n \leq 1009$ it enumerates all $(n-1)^3$ states, so the probabilities it returns are exact. Every assertion below is a restatement of a theorem in the body, and all of them pass.

```
import numpy as np, math

def enumerate_states(n):
    """Exhaustive sweep over the state space S of the structure lemma: (zeta, xi, u)."""
    v = np.arange(1, n, dtype=np.int64)
    Z, X = np.meshgrid(v, v, indexing='ij')
    Z, X = Z.ravel(), X.ravel()
    S = (Z + X) % n
    keep = S != 0
    Z, X, S = Z[keep], X[keep], S[keep]

    sig = Z + X # true integer sum
    nE = int(np.count_nonzero((sig % 2 == 0) &
                              (np.abs(Z - X) == sig // 2 + 1)))
    nE_line = int(np.count_nonzero((Z == 3*X + 2) | (X == 3*Z + 2))) # E structure thm
    assert nE == nE_line == 2 * ((n - 3) // 3) # exact E count

    q = a_ct = b_ct = 0
    for u in range(1, n):
        ZR = (Z * u) % n
        m = ZR > S
        z_, x_, s_, zr_ = Z[m], X[m], S[m], ZR[m]
        d = zr_ - s_
        a = s_ % d
        g = (a > 0) & (a != s_) & (a != zr_) # qualification
        z_, x_, s_, zr_, a = z_[g], x_[g], s_[g], zr_[g], a[g]
        q += a.size
        a_ct += int(np.count_nonzero(z_ == a)) # class A
        t = s_ + zr_
```

```
        both = (z_ % a == 0) & (t % x_ == 0) # class B
        if both.any():
            b_ct += int(np.count_nonzero(
                z_[both] // a[both] == t[both] // x_[both]))

    tot = (n - 1) ** 3
    assert abs(a_ct/tot - (q/tot)/(n-2)) < 1e-15 # exact A frequency
    return dict(n=n, PQ=q/tot, PA=a_ct/tot, PE=nE/(n-1)**2, PB=b_ct/tot)

for n in (101, 151, 251, 401, 503, 751, 1009):
    r = enumerate_states(n)
    print(f"n={r['n']:5d} Pr[Q]={r['PQ']:.6f} Pr[A]={r['PA']:.6e} "
          f"Pr[E]={r['PE']:.6e} Pr[B]*n^2/ln n={r['PB']*n*n/math.log(n):.3f}")

# --- secp256k1 ------------------------------------------------------------
from decimal import Decimal as D, getcontext
getcontext().prec = 60
n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141
PA = D(1) / 4 / (n - 2)
PE = D(2 * ((n - 3) // 3)) / D((n - 1) ** 2)
PB = D("0.19") * D(n).ln() / (D(n) * D(n))
print("Pr[A] = %.6E Pr[E] = %.6E Pr[B] = %.3E 1/n = %.6E"
      % (PA, PE, PB, D(1) / n))
```

B Notation

n	prime group order
$\zeta = s_{zk} = zk^{-1},\ \xi = s_{rxk} = rxk^{-1}$	hidden components (Definition 1)
$\sigma = \zeta + \xi \in \{s, s+n\}$	integer sum (2)
$s_{zr} = zr,\ d = s_{zr} - s,\ a = s \bmod d$	public auxiliaries (Definition 2)
$m_1 = \zeta/a,\ m_2 = (s + s_{zr})/\xi$	divisibility ratios, taken in $\mathbb{Q}$
$\mathcal{Q}$	qualification event (Definition 9)
$C,\ \mathbb{E}\|C\|$	candidate set of a rule; its expected size
$\tau(\cdot),\ \gamma$	divisor function; Euler–Mascheroni constant

All modular arithmetic is modulo n unless a relation is explicitly stated to hold in $\mathbb{Z}$.